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TFKAN: Transformer based on Kolmogorov–Arnold Networks for Intrusion Detection in IoT environment

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ABSTRACT

This work proposes a novel Transformer based on the Kolmogorov–Arnold Network (TFKAN) model for Intrusion Detection Systems (IDS) in the IoT environment. The TFKAN Transformer is developed by implementing the Kolmogorov–Arnold Networks (KANs) layers instead of the Multi-Layer Perceptrons (MLP) layers. Unlike the MLPs feed-forward layer, KAN layers have no fixed weights but use learnable univariate function components, enabling a more compact representation. This means a KAN can achieve comparable performance with fewer trainable parameters than a larger MLP. The RT-IoT2022, IoT23, and CICIOT2023 datasets were used in the evaluation process. The proposed TFKAN Transformer outperforms and obtains higher accuracy scores of 99.96%, 98.43%, and 99.27% on the RT-IoT2022, IoT23, and CICIOT2023 datasets, respectively. The results indicate that the developed Transformer using KAN shows promising performance in IDS within IoT environments compared to MLP layers. Transformers based on KANs are on average 78% lighter, in parameter count, than Transformers using MLPs. This makes KANs promising to be a replacement for MLPs.

1. Introduction

The fast evolution of the Internet of Things (IoT) has accelerated the broad implementation of several smart city applications. Today, the easy spreading of devices in smart home systems, various Internet of Vehicles (IoV) parts, and industrial machinery into the Internet is an irreversible tendency. However, the resource-challenged nature of IoT hardware and the insufficiency of security infrastructure make these devices easy targets for cyber intrusions [1]. Due to the low cost, low power, and limited computational capability features of IoT devices, there arise significant problems when trying to implement a suite of strong security mechanisms, including encryption algorithms [2]. Moreover, the inherent openness of IoT ecosystems leads to the devices, networks, and platforms remaining easily exposed to attack or unauthorized control. Since these gadgets are directly connected to the internet, other networks, and local area networks, hackers may launch extensive attacks by extracting private and sensitive data from devices that are identical to them using just rudimentary tools.

In this regard, the vulnerability of separate IoT devices could easily become a critical point of exploitation, resulting in a major threat to

overall IoT security. Cyber attackers can exploit these security flaws to tamper with software, making the devices into zombie machines. This brings along devastating impacts, such as loss of data, system malfunction, and the leaking of highly personal information [3]. Owing to the lack of proper security, most IoT devices fall victim to network-layer issues [4]. The most devastating of these comes in the form of Denial of Service (DoS) and botnet attacks, which, given their widespread nature, have an element of stealth. These attacks, as a result, make it possible for IoT devices to perform in an abnormal manner, giving way to the possible takeover of devices by attackers, and setting the stage for devastating situations. How, in such a scenario, an IDS deployed in the IoT network can act as a front-line defense. A Machine Learning (ML) based intrusion detection system (IDS) in particular may be very useful in analyzing the traffic patterns of networks and spotting ideas that have malicious intent. Methodologically speaking, IDS may be divided into two categories: anomaly-based detection and signature-based detection. Each category relies on a distinct type of detection.

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The development of ML technology established significant advancement in technologies for Intrusion Detection Systems in recent years, greatly improving the efficiency and accuracy of detection [5]. The IDS widely uses Supervised Learning algorithms, for example, Gradient Boosting Decision Trees (GBDT), the Support Vector Machine (SVM), and eXtreme Gradient Boosting (XGBoost), but there are still many limitations restricting them: SVM does not work well when handling training on Big dataset and multi classification tasks. The XGBoost model cannot keep large-dimensional feature data effectively, also, the GBDT model cannot flexibly prevent overfitting problems and ignore feature correlations. Therefore, this method combines multiple algorithms to take full advantage of their characteristics and achieve the best results. Deep learning (DL) technologies, including Convolutional Neural Networks (CNN) and Long Short-Term Memory (LSTM) networks, have been applied to IDS, but CNN is best at space-aware data learning, while LSTM is good at time-aware data. Although they have their shortcomings – the weak sensitivity of CNN enables the feature to aggregate over space instead of time, and the LSTM has a longer processing time – combining them in a CNN-LSTM framework is an effective way to consider the features of both spatial and temporal data in an effective and powerful method to detect malicious traffic [6].

Transformers, known for their ability to model dependencies that exist across very large distances, have recently been applied in IDS within the IoT. Specially, these models, especially those that work on an attention basis, process large volumes of data, model it for patterns, and have high accuracy in detecting anomalies. For example, transfer the attention mechanism to the Modified Transformer Neural Network to advance in detection accuracy of various kinds of attacks, with considerable improvements in accuracy, precision, recall, and F-score metrics, compared to the LSTM and RNN models [7]. A second example is the Multiple Transformers Single Estimator (MTSE) Learning pipeline to optimize feature sets and ML algorithms for accuracy in the recognition of IoT devices [8]. However, applying transformers to IDS in the IoT also has significant drawbacks. The large amount of computing resources [9] and memory requirement of transformers is a big challenge, especially for resource-constrained devices, like those in IoT. These drawbacks of the transformers have been addressed by exploring techniques such as transformer pruning to reduce the size of the model and the complexity of its inference without affecting its performance level. Even with such optimizations, obtaining real-time processing and maintaining efficiency in the widely varying heterogeneous settings of IoT has been very challenging. Additionally, the often-considerable amount of data that transformers usually need for training is not always available; in some cases, the datasets are imbalanced, leading to biases in detections [10]. Thus, while these approaches seem promising, further research is warranted explicitly to mitigate such drawbacks and to enhance the practicality and effectiveness of transformers and related models in highly dynamic, heterogeneous IoT spaces.

Kolmogorov–Arnold Networks (KANs) efficiently approximate high dimensional functions through the decomposition of such functions into sums of univariate functions, a theoretical framework boosting both computational efficiency and expressiveness [11]. In this regard, they are particularly good at capturing complex, non-linear data interactions. The most common areas in which these layers are applied include those where complex functionalities must be approximated, and hence, KAN layers in place of conventional Multi-Layer Perceptrons (MLPs) have significant advantages. Thus, KAN layers propose a theoretical method to decompose high-dimensional functions in sums of simpler ones, namely univariate functions—thus enhancing computational efficiency and expressiveness on the one hand, and are particularly good for modeling long-range dependencies and non-linear interactions on the other. Most importantly, KAN layers in transformers are promising to achieve better performance in tasks that require a higher level of pattern recognition and complex decision-making from the dataset, thus allowing models to achieve better performance for the corresponding

problem requirements and further benchmarking the potential of the current architectures in comparison to classical MLP layers.

The work proposes a novel Transformer model based on KANs layers (TFKAN) as a replacement of MLP layers for IDS in an IoT environment. Transformers use MLPs for processing, which can be limited in capturing complex, high-dimensional patterns inherent in the datasets. Unlike standard MLP feed-forward layers, KAN layers have dynamic weights and implement learnable univariate function, this produces a more compact representation. In other words, KAN can achieve high accuracy and performance with fewer trainable parameters than a larger MLP [12]. In an IoT setting with many low-power devices, a lighter model is advantageous. By leveraging KAN layers, which have a strong theoretical foundation in function approximation [11], our TFKAN model reduces the parameter count while preserving accuracy making deployment on resource-limited devices more feasible. The main contribution of this paper can be summarized as follows:

- Proposed a novel Transformer model based on KANs layers, termed TFKAN. It integrates a Transformer model with KAN layers as a replacement for MLP layers.
- Implementing the proposed TFKAN Transformer for IDS with IoT environment for both Binary and Multi classifications tasks.
- Evaluation and assessments proposed TFKAN on the most recent IOT datasets such as RT_IOT2022 [13], IoT23 [14] and CICIOT2023 [15] datasets.

The rest of the paper is organized as follows: Section 2 reviews related work for the main topics of this paper. Section 3 presents the background of Transformers and KANs, following Section 4 proposes a clear explanation of the suggested TFKAN Transformer. Section 5 presents the evaluation results of our conducted experiments, an outline of the utilized datasets, performance, and comparative results. Lastly, our work concluded and future trends of this work were summarized in the last Section 6.

2. Related work

McHugh et al. [16] laid the foundation for IDS research, emphasizing IDS as part of a defense-in-depth strategy for organizations. They provided guidelines for deploying, operating, and maintaining IDS, underlining that an IDS must continuously monitor all events to react to suspicious activities. Building on this foundation, Verwoerd and Hunt [17] discussed various IDS methodologies and identified normal and abusive system behavior: statistical models, neural networks, and genetic algorithms. It became a good source for an exhausting view of IDS technologies at that time.

In a review by [18], the authors analyze some of the IDS devices as well as software and illustrate the main categories and performance measurements. This paper reviews how IDS monitors network traffic for malicious activities that violate the normal functionality of the system and therefore cause policy violations. The review gives a broad picture of IDS technology, its applications, etc.

The review in [19] includes works related to the application of DL in IDS, covering 23 papers based on different approaches such as computers and vision, NLP, and voice recognition. This review reminded us at how automated an intrusion detection system needs to be, in light of the ever-changing nature of cyber threats. It is noted that DL is one way of making real-time intrusion detection better. In comparison with traditional approaches, an improvement in performance with DL is pointed out for intrusion detection. Another one [20] developed an intelligent DL model using TensorFlow for intrusion detection with features focusing on improvement of the detection rate and reduction of false positives, especially for small sample data. It proposes a big data storage solution based on Hadoop and constructs knowledge of

intrusion feature behavior in a model base. On the other hand, decision trees are used to match architected strategies for implementing automatic security control, and experiments showed that they considerably improved the detection accuracy and had the capability to differentiate between known and unknown intrusions.

Further development in IDS has been made. One study [21] discusses the development of an efficacious and intelligent IDS employing DL approaches to enhance cyber threat detection. They reinforce deep neural networks like CNN, RNN, DBN to learn and extract temporal patterns in network data alias automatic identification of known and novel threats by catching attack signatures as well as abnormal behaviors. Strategy-wise, these are transfer learning and dimensionality reduction that work on the issues of feature extraction, associated scalability, and interpretability, therefore providing a higher detection rate and lower false positives via exhaustive experiments. Authors in [22] proposed a novel intrusion detection architecture that consists of a CNN module, an LSTM module, and an SVM classification function. The approach intrinsically addresses all the shortcomings associated with the presently used methods since it considers the emerging spatial relationships and long-term dependencies in the case of the given dataset. The comparisons reflect the excellence of the proposed multi-class detection system over traditional ML techniques.

Numerous Hybrid IDS (HIDS) have been suggested in the literature to improve cyberattack detection. A HIDS integrating signature-based and anomaly-based methodologies was introduced by [23] to identify DDoS assaults. Multidimensional Gaussian Mixture Models (GMMs) were used for anomaly-based detection, while SNORT was used for the signature-based component. Similar to this, [24] created a hybrid IDS that could identify both known and unidentified threats by combining Network Traffic Anomaly Detection (NETAD) and Packet Header Anomaly Detection (PHAD). [25] suggested a mixed IDS for the smart grid system that learns temporal state-based patterns to detect cyberattacks in power system situations ranging from cyber disruptions to normal operations.

Authors in [26] presented a Multitiered Hybrid IDS (MTH-IDS), that employs many ML models to determine assaults on both in-vehicle and external networks, which is one example of a more recent development. Using distinct CL-KMeans based on K-Means, MTH-IDS employs the CICIDS2017 and CAN-intrusion datasets for evaluation. This helps it identify zero-day attacks. [27] introduced a new hybrid IDS for classifying IoT attacks. It examined its usage of a C5 classifier and One-Class SVM with the Bot-IoT dataset. Furthermore, in order to detect unknown DoS/DDoS attacks in IoT networks, [28] introduced a semi-supervised hybrid model that tests on the Bot-IoT, CIC-IDS-2017, and CIC-IDS-2018 datasets using nearest-neighbor ensembles (iNNE).

This paper [29] developed a new IDS based on ML that minimized identity risks in IoT space. The developed system uses sophisticated algorithms to calculate permission and credibility scores of accessing entities, duly enabling an accuracy of 94.7% for the purpose of identification, avoiding invasive access by filtration. In addition, [30] addressed the challenge of this imbalanced data in IoT environments by proposing TMG-GAN, a multi-generator GAN structure generating diverse attack data, thereby bringing about great improvements in IDS detection capabilities through advanced data augmentation techniques. This paper [31] introduced the BR-HIDF framework, an anti-sparsity host IDS that implements multi-granularity feature extraction to reduce feature space dimensionality and result in an increase in the detection's accuracy, which makes it quite superior in terms of measurement metrics like accuracy, precision, recall, and F1 score. [32] have designed an SDN-based VANETs defense against DDoS attacks built on the TabNet model with improved results against traditional ML techniques, whose detection rate could reach 99.42% percent. Their work presents an exhaustive analysis and enhancement of TabNet that puts this contribution in a good position in the state of art about protection against cyber-attacks on Intelligent Transport Systems. Finally, [33] proposed a framework of multi-label classification based on sequential methods

to improve the efficiency of IDS. This method trains on multiple sets of labels to avoid noise or interference, it keeps all the original labels, and also it is representing relations between the labels without any increase in resource consumption as it was demonstrated in its application to data from the real world.

Recent studies have focused on enhancing IDS using transformer models, showcasing their effectiveness in identifying and mitigating cyber threats in various contexts. One study proposes a robust transformer-based IDS that uses a variant stacked encoder-decoder neural network with a self-attention mechanism to classify network traffic types, achieving high detection rates on the CICIDS2017 and CIC-DDoS2019 datasets [34]. Another approach leverages a sparse cross-attention-based transformer detector for intrusion object detection in overhead power systems, demonstrating superior efficient in noisy circumstances when compared to classical methods [35]. Further, a model combining edge computing and transformers for wireless sensor networks (WSN) intrusion detection showed significant improvements in accuracy by integrating k-Nearest Neighbor (kNN) and arithmetic optimization algorithms, tailored for detecting DoS attacks [36]. Additionally, a study on maintainable log datasets for evaluating IDS highlights the utility of transformers in simulating normal user behavior and multi-step attacks, providing a scalable solution for IDS benchmarking [37]. Lastly, the integration of transformers in multi-layer-controlled power electronics-dominated grids (PEDG) for IDS enhances situational understanding against hostile groups-points by leveraging mathematical theories and morphism techniques, ensuring grid stability and security [38]. These advancements illustrate the potential of transformer models in developing more sophisticated and accurate IDS frameworks. Table 1 summarizes these related works.

Recently, Authors in [12] developed a new DL model based on CNN and KANs called CKAN for IDS by replacing MLPs layers in CNN with KANs. They evaluated the CKAN model by using NSL_KDD, CICIOT2023, and TONIOT datasets. Their work proved the effectiveness of implementing the DL models based on KANs in increasing detection accuracy with lightweight models. In This work, we aim to build a Transformer based on KANs for IDS to increase the detection accuracy of intrusions in IoT environment.

3. Background

3.1. Transformers

Transformers are multi-layer neural networks designed by piling multiple encoder and/or decoder blocks, that employ the attention scheme described in the subsequent sections.

3.1.1. Attention

The likeness between each individual input token – which is represented as a vector of word embeddings – is calculated by the attention mechanism. Each input embedding in a basic Transformer design may often play one of three roles: Three input tokens are compared to each other: (1) query (Q)-the recent direction of the attention method; (2) key (K)-the input token that needs to be compared with the query; and (3) value (V) for calculating the output of the attention mechanism. To provide an output, the attention process directs a query to a group of key-value couples [39].

We denote the input $\mathbf{X} \in \mathbb{R}^{n \times d}$ with a series of n tokens with an embedding dimension of d . We linearly transform the input series $\mathbf{X}$ to query $\mathbf{Q}$, key $\mathbf{K}$, and value $\mathbf{V}$ based on the following equations:

$$\begin{aligned} \mathbf{Q} &= \mathbf{X} \cdot \mathbf{W}_q \\ \mathbf{K} &= \mathbf{X} \cdot \mathbf{W}_k \\ \mathbf{V} &= \mathbf{X} \cdot \mathbf{W}_v \end{aligned} \quad (1)$$

Table 1
Related work summary.

Paper	Model used	Advantages	Disadvantages
McHugh et al. (2000) [16]	IDS	Established foundational concepts in IDS	Limited by the technology and understanding at the time
Jacob and Wanjala (2017) [18]	Review of IDS	Provides a broad review of IDS developments	Lack of in-depth analysis of newer techniques
Shivhare et al. (2023) [22]	DL	Effective in various intrusion scenarios	High computational cost
Vadursi et al. (2016) [23]	Anomaly Detection	Effective for anomaly detection	May produce false positives
Yang et al. (2021) [26]	Multi-Tiered Hybrid IDS	High detection accuracy	Complex system architecture
Khraisat et al. (2019) [27]	Hybrid IDS	Combines multiple techniques for better accuracy	Increased system complexity
Aydin et al. (2009) [24]	Hybrid IDS	Improved detection rates	Higher computational demands
Pan et al. (2015) [25]	Data Mining	Effective in power systems	Specific to power systems
Nguyen and Le (2023) [28]	Hybrid Learning Model	Robust against unknown attacks	High complexity
Zhang et al. (2023) [29]	Permission and Credibility-Based IDS	Effective in IoT environments	Specific to IoT, may not generalize
Ding et al. (2023) [30]	TMG-GAN	Handles imbalanced data well	Complexity in implementation
Thuan et al. (2024) [31]	Backbone Search for Object Detection	Effective in specific applications	Limited to object detection scenarios
Setitra and Fan (2024) [32]	Optimized TabNet	High accuracy in detecting DDoS attacks	May not be effective for other attack types
Hallaji et al. (2024) [33]	Ensemble-Based Multi-Label Classification	Expands analytical capabilities	Increased complexity in model design
Abd Elaziz et al. (2024) [12]	CKAN	High Detection accuracy	high processing time

such that the weight matrices W_q , W_k , and W_v are utilized to receive the Q , K , and V matrices. The scaled dot product attention is then computed using the Q , K , and V matrices, as seen below:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{Q \cdot K^T}{\sqrt{d_k}}\right) \cdot V \quad (2)$$

Following the scaled dot product operation between the query and key matrices, this algorithm uses a softmax function. Typically, the scale factor is $1/\sqrt{d_k}$, where d_k is the key dimension. This helps to reduce numerical instability and the vanishing gradient problem.

3.1.2. Types of attention mechanisms in transformers

Transformers use a variety of attention methods to improve their sequence data handling capabilities. These consist of cross-attention, veiled self-attention, and self-attention, each with a specific function in the design of the model.

- **Attention to Self:** The attention scores are determined by this process in the same order. It is an essential part of the encoder for the Transformer. In self-attention, the dimensions for query, key, and value are identical and are represented as $d_k = d_q = d_v$.
- **Masked Attention to Oneself:** vital for sequence prediction jobs, like machine translation, where the goal is to forecast future outputs by using tokens from the past. The model is shielded from further tokens in the sequence by a mask. The mask is involved to the scaled dot product of the query and key. It is represented as an upper triangular matrix M of size $n \times n$ (where n is the number of tokens). This is how the veiled attention is calculated:

$$\text{Masked Attention}(Q, K, V) = \text{softmax}\left(\frac{Q \cdot K^T}{\sqrt{d_k}} + M\right) V. \quad (3)$$

- **Cross-Attention:** allows tokens from two different sequences to interact with each other. Cross-attention is implemented in the Transformer decoder using queries from the last masked self-attention layer, where the last encoder provides keys and values. The keys and values are taken from the input series that the encoder processes and the queries match the intended output sequence.
- **Multi-Head Focus:** It has been demonstrated that adding several attention operations greatly improves a model's capacity to identify different kinds of interactions inside the sequence [39]. The

weight matrices W_q^i , W_k^i , and W_v^i , where i indicates the particular head, are used by each attention head independently. The outputs from each head ($\mathcal{H}$) are combined using the Transformer model and linearly transformed using a parameter matrix W_o to fit the model's dimension (see Fig. 1).

3.1.3. Position-wise feed-forward network

Each token in the sequence is processed by a two-layer feedforward network (FFN) following the output from the attention modules. Every token in the sequence undergoes an autonomous position-wise linear transformation. The network's parameters are uniform throughout all places in the sequence.

Let the multi-head attention block's output be $\mathcal{H}$ and the model dimension be d . $\mathcal{H}$ is converted by the first linear layer from dimension d to an middle dimension d_f , also called the feedforward dimension. The output of d_f is subsequently converted by the second linear layer back to the actual model dimension d . The FFN is defined as:

$$\mathcal{F}(\mathcal{H}) = \text{ReLU}(\mathcal{H} \cdot W_1 + b_1) \cdot W_2 + b_2 \quad (4)$$

The intermediate dimension d_f is typically larger than d .

3.1.4. Layer normalization and residual connections

Gradients can flow around non-linear activation functions with residual connections, and then layer normalization. By scaling all hidden layer values to extremely comparable ranges, layer normalization helps to prevent issues with gradients that explode or disappear as a result of repeated multiplication operations.

3.1.5. Encodings for positions

The self-attention module lacks intrinsic knowledge of the sequentiality of the tokens as it analyzes each token in turn in parallel. To indicate the exact location of each token, PE is employed for this specific purpose. These PE vectors are produced by transformers as a function of sine and cosine at various frequencies:

$$PE_{(\text{pos}, i)} = \begin{cases} \sin(\text{pos} \cdot \omega_k), & \text{if } i = 2k \\ \cos(\text{pos} \cdot \omega_k), & \text{if } i = 2k + 1 \end{cases} \quad (5)$$

$$\omega_k = \frac{1}{10000^{2k/d}}, \quad k = 1, 2, \dots, \frac{d}{2}$$

These PE have the same dimensions as the embeddings since they are summed with each input token's embeddings. The transformers model

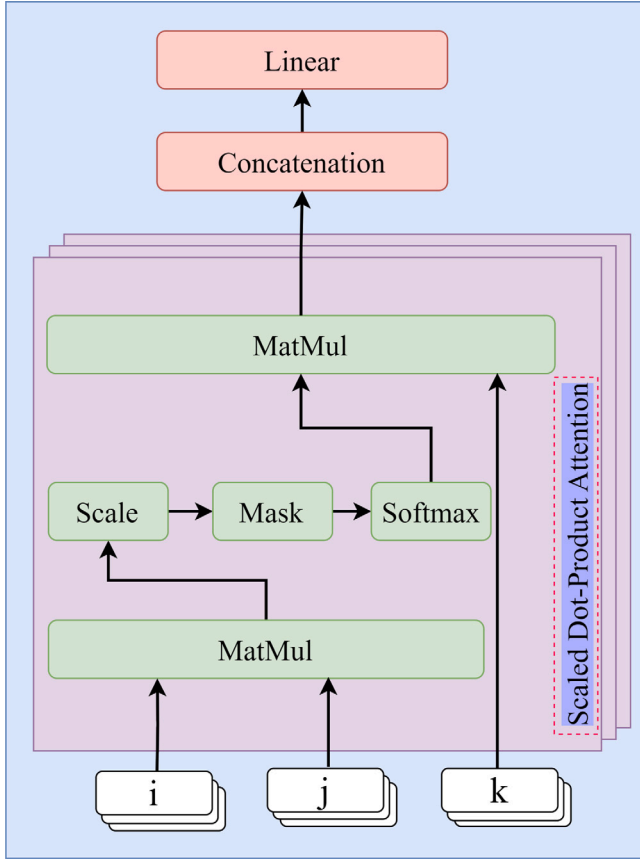


Fig. 1. Multi-head attention in the encoder and decoder, showcasing combined outputs from several attention heads.

captures interactions between elements that are either close or far in a sequence using the PE approach.

3.1.6. Transformer model usage

In public cases, transformer architectures fall into one of three types:

1. **Encoder-Only:** Tabular categorization was a suitable application for this design. It models the input sequence using only the encoder blocks. The input sequence is represented contextually in the output.
2. **Decoder-Only:** Language modeling and picture captioning both made use of this framework. It is used for activities requiring the creation of sequences and just requires the decoder blocks.
3. **Encoder-Decoder:** This design, which included many encoders and decoders, was applied to sequence-to-sequence modeling applications, including machine translation and text production.

3.2. Kolmogorov-Arnold networks (KANs)

Multi-Layer Perceptrons (MLPs) are based on the universal approximation theorem [40], which they developed from the original perceptron by Rosenblatt et al. (1958). According to this theorem, continuous functions may be represented by a feed-forward network with a single hidden layer inside restricted areas of $\mathbb{R}^n$. Kolmogorov-Arnold Networks (KANs) on the other hand, utilize the Kolmogorov-Arnold representation theorem [41], which characterizes every multivariate continuous function as the sum of univariate functions:

$$f(x_1, \dots, x_n) = \sum_{q=1}^{2n+1} \Phi_q \left(\sum_{p=1}^n \phi_{q,p}(x_p) \right), \quad (6)$$

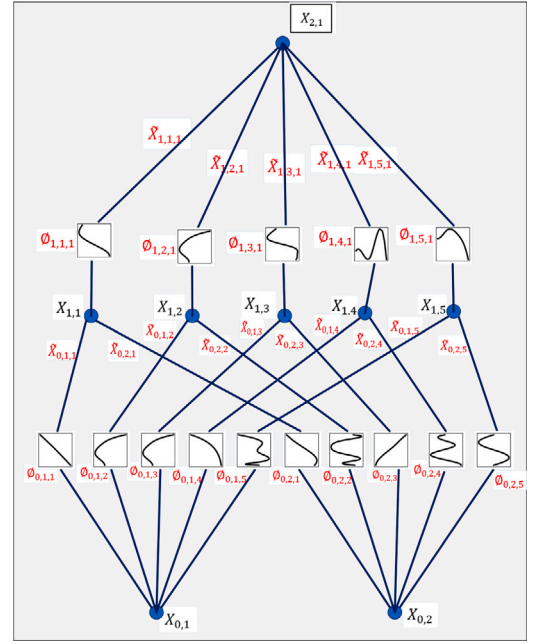


Fig. 2. Spline Activation Function of KANs.

where Φ_q are real-valued functions and $\phi_{q,p}$ translates each input x_p as a univariate function.

KANs define a layer as a set of mappable functions $\Phi = \{\phi_{q,p}\}$ with learnable parameters, structuring inner functions to construct complicated layers, whereas MLPs use layers of linear transformations followed by nonlinear activations. By grouping these functions into two sets of KAN layers, the richness of these functions is captured, providing a more sophisticated and subtle composition of functional mappings:

$$\Phi = \{\phi_{q,p}\}, \quad p = 1, 2, \dots, n_{in}, \quad q = 1, 2, \dots, n_{out} \quad (7)$$

These layers improve the modeling capabilities as well as the interpretability of the learning process by methodically dissecting the network's actions into smaller, more manageable parts. As seen in Fig. 2, the complex interactions between neurons across layers are therefore portrayed in an organized way, enabling a thorough comprehension of the neural network's performance.

In matrix form, this becomes:

$$\mathbf{x}_{l+1} = \underbrace{\begin{pmatrix} \phi_{l,1,1}(\cdot) & \phi_{l,1,2}(\cdot) & \dots & \phi_{l,1,n_l}(\cdot) \\ \phi_{l,2,1}(\cdot) & \phi_{l,2,2}(\cdot) & \dots & \phi_{l,2,n_l}(\cdot) \\ \vdots & \vdots & \ddots & \vdots \\ \phi_{l,n_{l+1},1}(\cdot) & \phi_{l,n_{l+1},2}(\cdot) & \dots & \phi_{l,n_{l+1},n_l}(\cdot) \end{pmatrix}}_{\Phi_l} \mathbf{x}_l \quad (8)$$

where Φ_l is the function matrix for the l th KAN layer. A general KAN network is a composition of L layers: given an input vector $\mathbf{x}_0 \in \mathbb{R}^{n_0}$, the output of the KAN is:

$$\text{KAN}(\mathbf{x}) = (\Phi_{L-1} \circ \Phi_{L-2} \circ \dots \circ \Phi_1 \circ \Phi_0) \mathbf{x} \quad (9)$$

4. Proposed transformers based on Kolmogorov-Arnold networks (TFKAN) model

This work proposes building a new Transformer (TFKAN) based on KAN layers as a replacement for MLPs layers. We replaced any MLPs layers such as Linear and Dense layers by KANs layers in the entire architecture. This novel integration employs the unique advantages of KANs including their capability to memorize more complicated

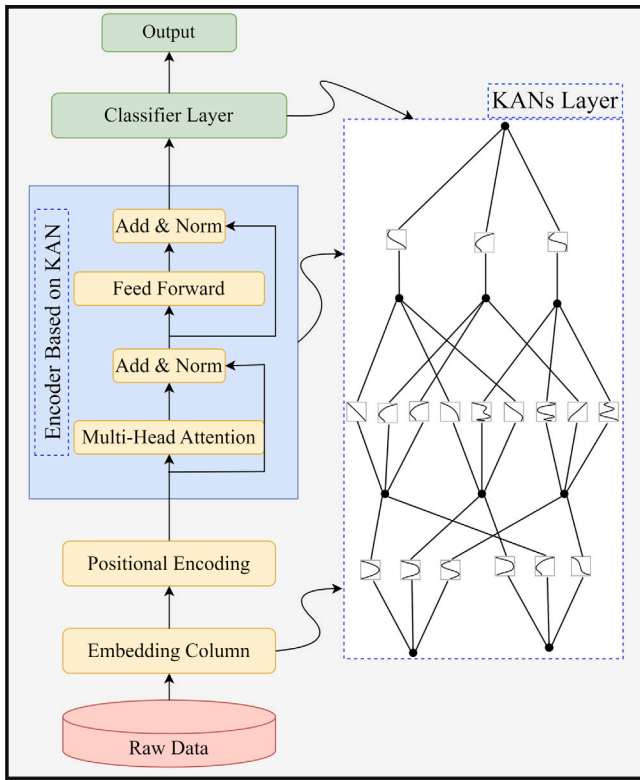


Fig. 3. Proposed TFKAN Model Flowchart.

function mappings with fewer parameters and improved interpretability. In , we introduced three architecture types for the usage of the Transformers, in this study, our target is tabular classification, so the proposed TFKAN model was implemented by using encoders only.

The proposed TFKAN model flowchart is presented in Fig. 3. It starts with reading IDS's dataset in raw tabular data form, the process begins by transforming the input into dense vector representations through an embedding column by utilizing KANs layer. This is a crucial transformation that makes it possible to have much richer feature representations. A positional encoding is added to the embeddings in order to preserve sequence order. This encoding allows the order of the rows to be appreciated by the model as it works through the data. Next is the Encoder layer, this layer implemented based on KANs layers by replacing any MLPs or Fully Connected Layers (FCLs) layers with KANs linear layers. It includes a series of operations, starts by multi-head attention, followed by Add and Norm layers, and a feed-forward. The multi-head attention permits to concentrate on other parts of the IDS's data simultaneously, capturing complex interdependencies. The encoder output is then followed by the classifier KANs layer and ends with the final output classifying according to the learned features as intruder/benign for binary classification, or the corresponding class in multi classification.

5. Experimental results

In the current section, we will demonstrate the testing performance of the TFKAN Transformers on recent datasets such as RT-IoT2022 [13], IoT23 [14], and CICIoT2023 [15]. We implemented a lighter version of TFKAN for simplicity and compared the results with a Basic Transformer (Utilizing MLPs layers) with the same architecture to achieve a fair comparison.

Table 2

Description of the RT-IoT2022 dataset.

Property	Value
Number of Classes	12 (9 attack types + 3 benign type)
Number of Samples	123,117
Number of Features	83

Table 3

Description of the IoT23 dataset.

Property	Value
Number of Classes	7 (6 attack classes + 1 benign type)
Number of Samples	48 003
Number of Features	24

5.1. Dataset overview

In this section, we will introduce a characterization of the datasets used in this study.

5.1.1. RT-IoT2022 dataset

The dataset RT-IoT2022 has been developed to be used in the UCI ML Repository [13]. It is an extensive dataset created to support research in real-time IoT environments, featuring a great number of IoT devices and including both normal and advanced network attack methodologies, therefore well-fitting the development and testing of IDS. The richness of the dataset in network traffic includes various elements within the bidirectional attributes, diversified adversarial behaviors, including Brute-Force SSH attacks, DDoS attacks, and Nmap scanning patterns. The data come from several IoT devices: ThingSpeak-LED, Wipro-Bulb, and MQTT-Temp, creating complex scenarios that could even be a challenge for network security systems. This assures the best and in-depth dataset analysis, since dataset is collected through Zeek, the network-monitoring tool, and the Flowmeter plugin develops the dataset. With over 123,000 instances and 83 features, the dataset forms a strong base for developing advanced security solutions to tackle the changing threat environment for IoT networks, as summarized in Table 2.

5.1.2. IoT23 dataset

The IoT23 dataset [14], produced by the Stratosphere IPS project, is thus a valuable resource on cybersecurity research in the IoT domain. There are more than 760 million packets and 325 million labeled flows real representative traffic, which are classified into both benign and malicious network activities for more than 500 hours. It includes many interactions of different IoT devices and various sophisticated cyber-attack simulations such as DDoS, brute force attacks, and scanning and thus provides rich, realistic environment to the development, testing and train the IDS. IoT23 is also designed to support development of various ML cultivations in Cyber security related issues. Correspondingly It provides well and realistic dataset that can support researches to evaluate the distinct security challenges of IoT networks. IoT23 is thus available for detailed security analysis and can also be much of a means in the research and development of new IDS solutions. The main statistics of IoT23 dataset were outlined in Table 3.

5.1.3. CICIoT2023 dataset

Presented by Neto et al. CICIoT2023 [15] is a dataset supporting the development and evaluation of IDS. This dataset is very detailed and provides a broad and practical testbed for assessing the effectiveness of security solutions tailored to the diverse range of IoT-specific cyber threats. It hosts network behavior from 105 IoT devices under attack, capturing the complexity and variability of such conditions with 46 distinctive features, including a total of 33 different attack types organized into seven classes, namely DoS, DDoS, Web-based, Reconnaissance, Spoofing, Mirai, and Brute Force, as summarized in Table 4.

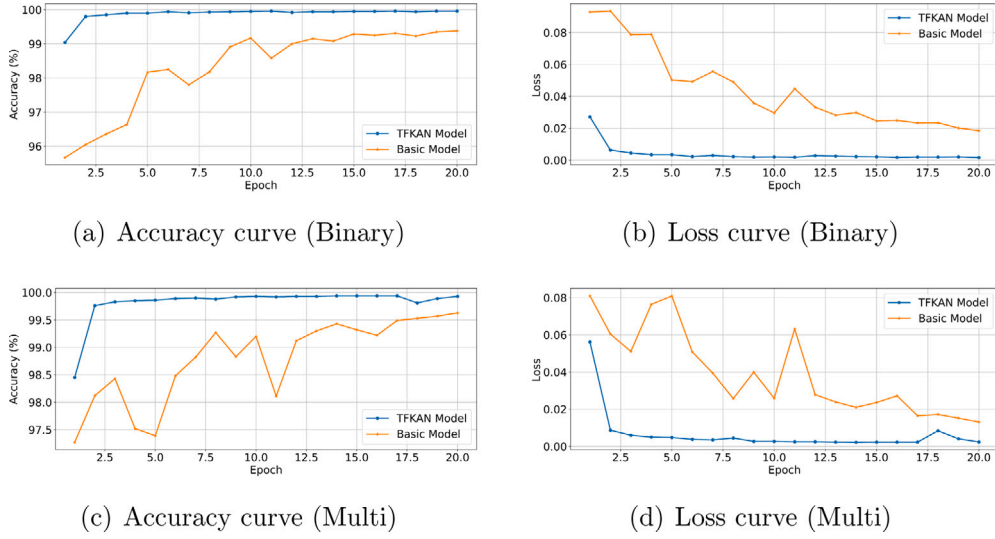


Fig. 4. The accuracy & loss curves of the proposed TFKAN Transformer compared to Basic Transformer on the RT-IoT2022 dataset for Both Binary & Multi Classification.

Table 4
Description of the CICIoT2023 dataset.

Property	Value
Number of Classes	9 (8 attack classes + 1 benign class)
Number of Samples	46,686,579
Number of Features	46

Table 5
Parameters setting for both transformers.

Parameter	Value
Batch Size	32
Learning Rate	0.001
Optimizer	Adam
Device	GPU
Epochs	20
embedded dim	64
number of heads	4
number of layers	2
dropout rate	0.5

5.2. Data preprocessing and experimental setup

This section presents the preprocessing details for each dataset to ensure clarity. The preprocessing steps applied to the RT-IoT2022, IoT23, and CICIoT2023 datasets are summarized as follows:

- **Data Cleaning:** Initial examination was conducted to eliminate instances containing missing or invalid values. This step,
- **Feature Selection:** Dropping the non-informative fields (e.g., flow IDs, timestamps).
- **Feature Encoding:** Categorical fields such as protocol type and others, were encoded by using one-hot encoder method.
- **Normalization:** All numeric features were normalized to [0, 1] range using min-max scaling.
- **Binary vs Multi-Class Labels:** For binary classification, we labeled all attack instances as “malicious” and benign instances as “benign”. For multi-class classification, we used the finer-grained attack categories as provided at each dataset.
- **Class Imbalance Handling:** We used the original class distributions without resampling, the evaluation metrics (precision, recall, F1) help ensure performance is judged beyond overall accuracy.
- **Train-Test Split:** For each dataset, we randomly split the data into 80% training and 20% testing sets (stratified by class).

All experiments were conducted on a Laptop with a Core i7 processor, Nvidia GTX 1650TI graphics card, and 32GB of RAM by using Python 3.12 and PyTorch library. All experiments were conducted to 20 epochs using Learning rate (0.001) and the other configuration settings such as the batch size assigned to 32, and the use of the Adam optimizer. The embedding dimension is 64, utilizing 4 attention heads and 2 layers in their architecture. Also, a dropout rate of 0.5 is used to control the overfitting during training. These settings for both models and are outlined in Table 5.

5.3. The evaluation metrics

In ML evaluation and Assessing methods, several key metrics are utilized to evaluate the significance of classification models described as follows:

- **Accuracy** measures the general correctness of the model and is defined by the equation as follows:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (10)$$

where TP , TN , FP , and FN represent the true positives, true negatives, false positives, and false negatives, respectively. It gives a widespread presentation of how the model is accurate across all classifications.

- **Precision** quantifies the accuracy of positive predictions and is calculated as follows:

$$\text{Precision} = \frac{TP}{TP + FP} \quad (11)$$

This metric is essential when the consequence of a false positive is meaningful.

- **Recall**, also known as sensitivity, measures the model’s capability to catch positive examples, formulated as follows:

$$\text{Recall} = \frac{TP}{TP + FN} \quad (12)$$

It is specifically important in procedures where ignoring a positive instance (i.e., false negative) assumes a richer penalty.

- **F1 Score** is the harmonic standard of Precision and Recall, expressed as follows:

$$\text{F1 Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (13)$$

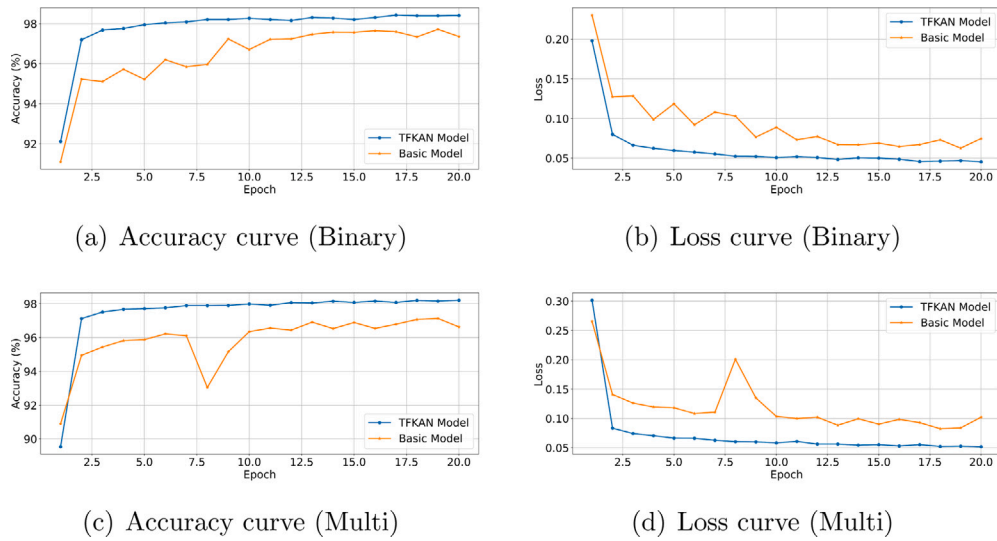


Fig. 5. The accuracy & loss curves of the proposed TFKAN Transformer compared to Basic Transformer on the IoT23 dataset for Both Binary & Multi Classification.

Table 6

Training performance of both transformers on the RT-IoT2022 dataset.

Model	Task	Params	Time (s)	Loss	Accuracy
Basic	Binary	1 205 058	522.17	0.0184	99.38%
	Multi	1 205 578	456.51	.0131	99.63%
TFKAN	Binary	169 472	1484.96	0.0016	99.96%
	Multi	174 592	1545.63	0.0022	99.94%

Table 7

Training performance of both transformers on the IoT23 dataset.

Model	Task	Params	Time (s)	Loss	Accuracy
Basic	Binary	1,197,762	177.65	0.0626	97.72%
	Multi	1,198,087	215.25	0.0824	97.13%
TFKAN	Binary	132,992	654.33	0.0454	98.43%
	Multi	136,192	662.56	0.0518	98.21%

This metric is valuable if the balance requires Precision and Recall, especially in cases of uneven class distribution.

5.4. Results and discussion

We discuss the constructed model with other models and show the results in this part. Tables 6–8 present the performance analysis of the Basic and TFKAN Transformers across the RT-IoT2022, IoT23, and CIIoT2023 datasets, respectively. In terms of Accuracy and loss, Table 6 presents the training results on the RT-IoT2022 dataset, in binary classification tasks, the TFKAN Transformer shows high accuracy (99.96%) with minimum loss metrics (0.0016). The TFKAN model has (169472) trainable parameters, whereas the Basic model has (1205058). Thus, TFKAN uses 75% fewer parameters than the baseline model in the same conditions. However, the Basic Transformers required less training time (522.17) seconds compared to TFKAN. Also, for multi classification, TFKAN achieved the most heightened accuracy 99.94% with minimum loss value (0.0022), and the number of trainable parameters (174592), i.e., It is lighter than its competitor by (75%).

When involved to the IoT23 dataset, as presented in Table 7, the TFKAN similarly holds the highest accuracy (98.43%) with the best least loss value (0.0454) and accuracy (98.21%) with loss (0.518) for binary and multi classification, respectively. TFKAN has a number of trainable parameters (132992) and (136192) for binary and multi classification, respectively. This means TFKAN is lighter than the Basic Transformer by (80%). Table 8 presents the training results for the CIIoT2023

Table 8

Training performance of both transformers on the CIIoT2023 dataset.

Model	Task	Params	Time (s)	Loss	Accuracy
Basic	Binary	1,200,450	8,698.83	0.0209	99.03%
	Multi	1,200,968	8,023.91	0.056	97.98%
TFKAN	Binary	146,432	23,260.56	0.0161	99.28%
	Multi	150,912	25,986.55	0.0288	98.96%

dataset. The TFKAN outperforms the Basic Transformer across accuracy (99.28%), loss (0.0161) and accuracy (98.96%), loss (0.0288) for both binary and multi classification, respectively. Also, the TFKAN Transformer has trainable parameters (146,432) and (150,912) for both binary and multi classification, respectively, this means that it is lighter than the Basic by (78%). It is worth mentioning that the TFKAN Transformer required the longest training time in all training processes across the three datasets compared to the Basic Transformer. From these results, it is concluded that TFKAN represents a robust method whose process could be expanded in the real world for better accurate and more elaborate IDS in IoT networks.

To clearly present the superiority of the TFKAN Transformer against the Basic Transformer, we plot the accuracy & loss curves for both Transformers. Figs. 4(a), 4(c), 5(a), 5(c), 6(a), and 6(c) show that accuracy curves in both binary & multi classification for RT-IoT2022, IoT23, CIIoT2023 datasets, respectively. The TFKAN Transformer achieves higher accuracy earlier in the training process and sustains it across epochs. The accuracy curves prove that the TFKAN (Blue color) has faster convergence and superior stability across epochs than the Basic Transformer (Orange color) for the three datasets. While Figs. 4(b), 4(d), 5(b), 5(d), 6(b), and 6(d) show the loss curves for both Binary and multi classification on RT-IoT2022, IoT23, CIIoT2023 datasets, respectively. The loss curves prove that the TFKAN achieved the lowest loss value with high stability.

Overall, Figs. 4–6 show that the TFKAN model has higher accuracy and less loss across all epochs when compared to binary or multi classification tasks. The TFKAN model converged very quickly, and maintenance of more increased accuracy corresponded to the Basic Transformer. Besides, it is found that the TFKAN model loss curve had a quick and stable convergence when compared to the Basic Transformer and generalized better. As observed, these curves show the efficient learning and robust performance of the proposed TFKAN Transformer.

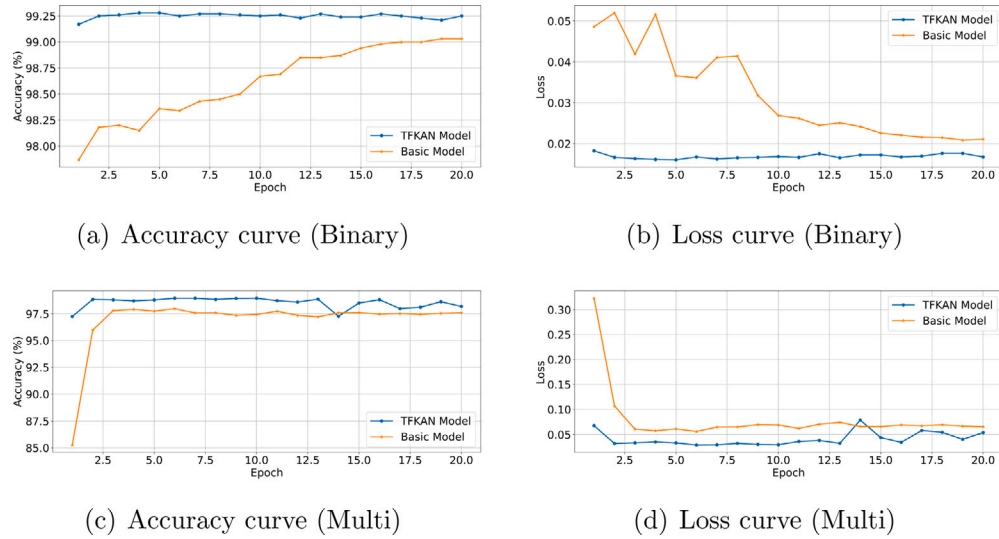


Fig. 6. The accuracy & loss curves of the proposed TFKAN Transformer compared to Basic Transformer on the CICIOT2023 dataset for Both Binary & Multi Classification.

Table 9

Performance of the proposed TFKAN Transformer for the RT-IoT2022 dataset compared to Basic Transformer.

Model	Binary Classification				Multi Classification			
	Acc.	Prec.	Recall	F1	Acc.	Prec.	Recall	F1
Basic	99.15	99.43	99.62	99.52	99.55	99.54	99.55	99.55
TFKAN	99.96	99.99	99.97	99.98	99.91	99.93	99.91	99.92

Table 10

Performance of the proposed TFKAN Transformer for the IoT23 dataset compared to Basic Transformer.

Model	Binary Classification				Multi Classification			
	Acc.	Prec.	Recall	F1	Acc.	Prec.	Recall	F1
Basic	98.08	97.15	98.67	97.90	97.03	97.18	97.03	97.07
TFKAN	98.43	97.32	99.26	98.28	98.35	98.31	98.35	98.26

Table 11

Performance of the proposed TFKAN Transformer for the CICIOT2023 dataset compared to Basic Transformer.

Model	Binary Classification				Multi Classification			
	Acc.	Prec.	Recall	F1	Acc.	Prec.	Recall	F1
Basic	99.11	99.56	99.53	99.54	97.84	97.86	97.84	97.74
TFKAN	99.27	99.71	99.55	99.63	98.85	98.81	98.85	98.78

5.5. Comparative results

Table 9 shows the performance metrics for both TFKAN and Basic Transformers on the RT-IoT2022 dataset on both binary and multi classification tasks, which were used to assess TFKAN's effectiveness using a testing set. The TFKAN model yielded the following results for binary classification: 99.96% accuracy, 99.99% precision, 99.97% recall, and 99.98% F1 score, which is better than the Basic Transformer. The TFKAN Transformer also achieved 99.91% accuracy, 99.93% precision, 99.91% recall, and a 99.92% F1 score in multi classification tasks. The consistency of these high metrics shows how the TFKAN model is reliable and strong against various types of network assaults.

Table 10 further depicts performance on the IoT23 dataset. Again, the overall performance of the TFKAN Transformer outperforms the competition in both binary & multi classification. In the case of binary classification, the TFKAN Transformer resulted in 98.43% of accuracy, 97.32% precision, 99.26% recall, 98.28% F1 score, outperforming the Basic Transformer. In multi classification cases, the TFKAN Transformer provided an accuracy of 98.35%, a precision of 98.31%, a recall of

98.35%, and an F1 score of 98.26%. Such uniformity in all these criteria reflects the robustness of the Transformer in distinguishing the cases of multi classification. These results indicate that the TFKAN Transformer can cope with a variety of complex attack types in the IoT23 dataset.

The performance on the largest dataset in the current study, CICIOT2023, is presented in Table 11. Once more, the TFKAN Transformer performs better than the competition across the board for both binary and multi classification tasks. The TFKAN Transformer yielded the following results for binary classification: 99.27% accuracy, 99.71% precision, 99.55% recall, and 99.63% F1 score. These outperform rival models. The TFKAN Transformer obtained 98.85% accuracy, 98.81% precision, 98.85% recall, and a 98.78% F1 score in multi classification tasks. The robustness of the Transformer in recognizing the multi classification scenarios is highlighted by this consistency across measures. The outcomes further show that the CICIOT2023 dataset's varied and complex assaults can be effectively handled by the TFKAN Transformer.

On the RT-IoT2022, IoT23, and CICIOT2023 datasets, respectively, the performance comparison of the proposed TFKAN Transformer to the Basic Transformer is shown in Figs. 7–9. The accuracy, precision,

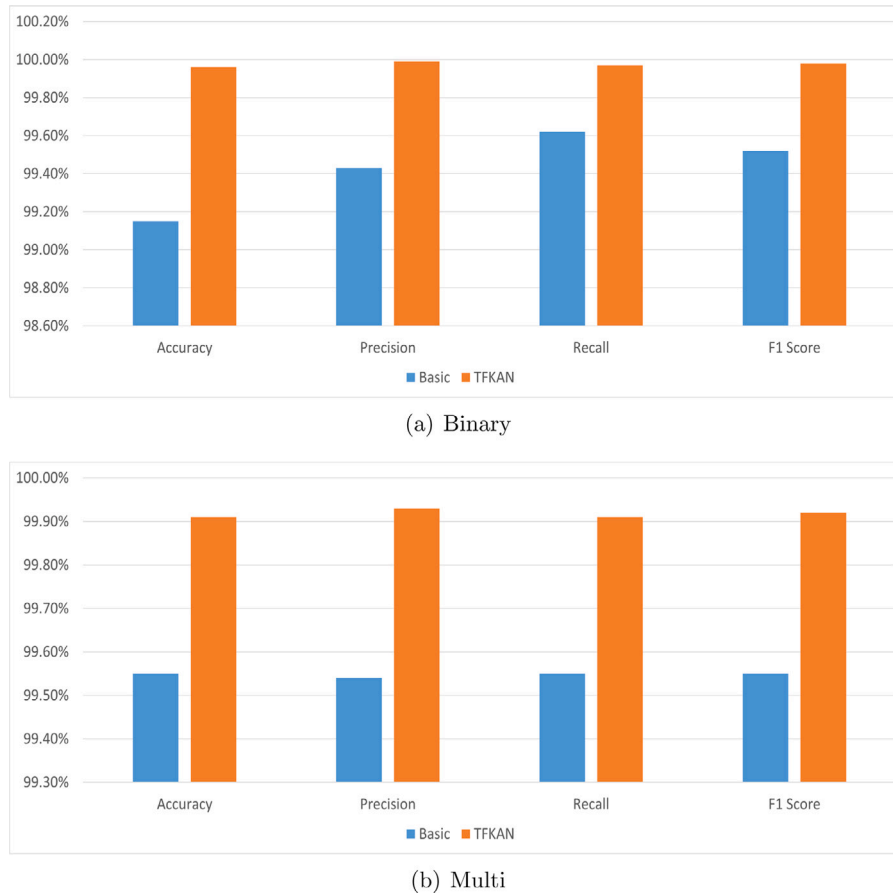


Fig. 7. The Bar chart of the Proposed TFKAN Transformer Compared to the Basic Transformer on RT-IoT2022 Dataset.

recall, and F1 score metrics for binary and multi classification tasks are shown graphically in these Figures. The TFKAN Transformer consistently outperforms the Basic Transformer and achieves the highest values in all metrics across all three Figures. In binary classification, the TFKAN Transformer exhibits exceptional superiority in accuracy and recall together with better F1 scores (Figs. 7(a), 8(a), and 9(a)). In multi classification, the TFKAN Transformer stays ahead and shows generalizability across several datasets (Figs. 7(b), 8(b), and 9(b)). These graphic comparisons highlight how well the TFKAN model performs in improving IDS accuracy across a range of IoT scenarios.

In terms of Computational efficiency such as the model size and speed, the proposed TFKAN model demonstrates superior efficiency and accuracy for real-time IDS in IoT environments outperforming the Basic Transformers. As discussed previously, the TFKAN reduces the trainable parameter count by 75%–80% across datasets. For example, for the RT-IoT2022 dataset showing a reduction from 1.2M parameters in the Basic Transformer to 169K in TFKAN, this translating into an 85% reduction in memory consumption. This lightweight nature makes TFKAN suitable for deployment on IoT devices with limited computational resources. Despite requiring 2.8× longer training time due to the learnable activation functions in KANs, the model maintains fast inference performance. The TFKAN processes approximately 500 samples per second, which is only 60% of the Basic Transformer's 800 samples per second and also offers higher classification accuracy. These results indicate the benefits of TFKAN model and prove its efficient, making it ideal for scalable real-time IDS deployment. Additionally, the model's reduced memory footprint translates into lower energy consumption which is a crucial factor for battery-powered IoT nodes.

The presented Figs. 7–9 and Tables 9–11 show how well the proposed TFKAN model performs when compared to the Basic Transformer

(implemented by MLPs layers) on the RT-IoT2022, IoT23, and CICIoT2023 datasets. In binary and multi classification tests, the TFKAN Transformer consistently delivers superior accuracy, precision, recall, and F1 scores. Its effectiveness demonstrates how reliable it is in identifying IoT intrusions. This suggests that the KAN layers are a potent substitute for the MLP layers as the industry-leading IDS solution. Also, our experiments found that the proposed TFKAN Transformer needs more time for the training phase, this is also mentioned by the authors of KANs in [11]. This may be considered as one drawback of the proposed TFKAN.

5.6. Comparative with literature works

The comparative literature of recent models used for IDS based on the datasets used in this work is shown in Table 12. According to the Table and for the RT_IoT2022 dataset, the Autoencoder model proposed by [42] achieved very low accuracy (91.4%). The proposed TFKAN model achieved higher accuracy (99.96%) compared to literature models such as, SVM [43], CNN [44], and Logistic Regression (LR) [45] achieving an accuracy of (92.9%), (97.31%), and (96%), respectively. For the IoT23 dataset, the TFKAN model achieved the best performance in terms of accuracy (98.43%) while comparing it to literature models such as DNN [50], CNN-LSTM [49], LSTM & CNN [48], LR [47], and ML [51] which they achieved an accuracy of (93%), (95%), (95%), (96.1%), and (96.7%), respectively. Lastly, for the CICIoT2023 dataset, some literature showed low detection capabilities such as the TBFMD model [54] which achieved 79.07% of accuracy. However, some literature showed low performance compared to the proposed TFKAN (99.27%) of accuracy such as LIDS-MV [55], LTSM [53], DNN based on Federated Learning (FL) [52], and Convolutional Kolmogorov–Arnold

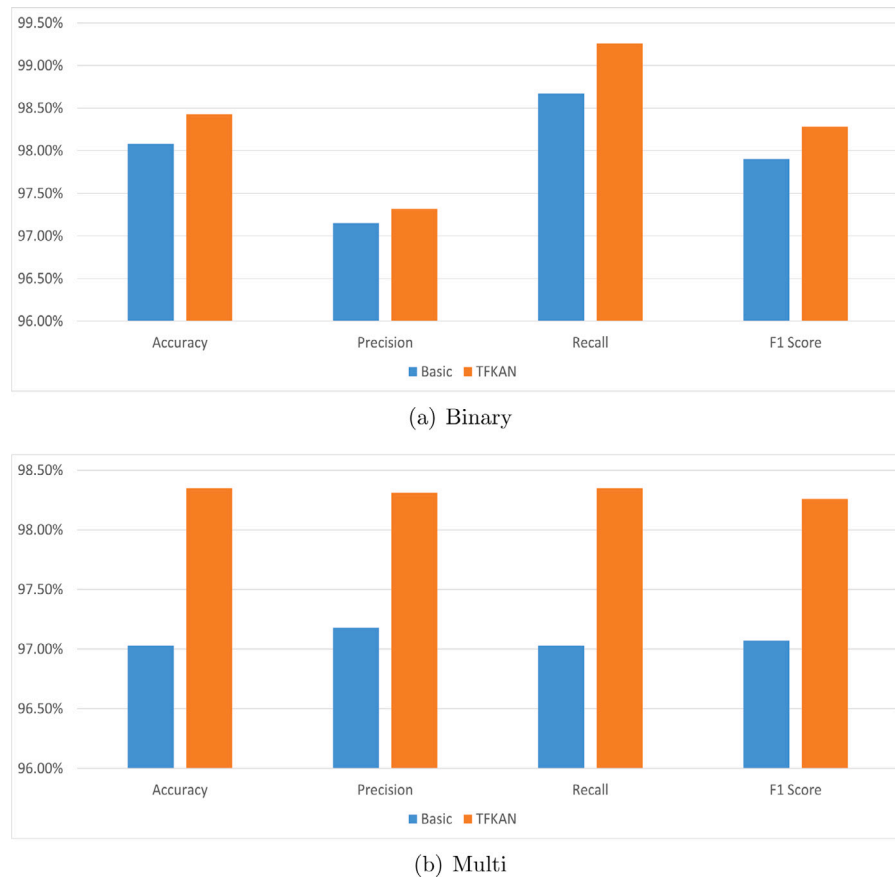


Fig. 8. The Bar chart of the Proposed TFKAN Transformer Compared to the Basic Transformer on IoT2022 Dataset.

Table 12

Comparative literature of recent models for IDS.

Dataset	Model	Accuracy (%)
RT_IoT2022	Autoencoder (2024) [42]	91.4
	SVM (2024) [43]	92.9
	CNN (2024) [44]	97.31
	Logistic Regression (2024) [45]	96
	Optimized TabNet (2025) [46]	99.16
	Proposed TFKAN	99.96
IoT23	Logistic regression (2021) [47]	96.1
	LSTM & CNN (2024) [48]	95
	CNN-LSTM (2024) [49]	95
	DNN (2023) [50]	93
	ML (2023) [51]	96.7
	Proposed TFKAN	98.43
CICIoT2023	DNN based on FL (2023) [52]	99
	LTSM (2023) [53]	98.75
	TBF-MD (2024) [54]	79.07
	CKAN (2024) [12]	99.22
	LIDS-MV (2024) [55]	98.35
	Proposed TFKAN	99.27

Networks(CKAN) [12] achieving accuracy of 98.35%, 98.75%, 99%, and 99.22%, respectively.

This literature ensures the potential of Transformer design based on KANs in detecting intrusions. The proposed TFKAN Transformer stands out by achieving the best accuracy of 99.96%, 98.43%, and 99.27% for the RT-IoT2022, IoT23, and CICIoT2023 datasets, respectively. This performance confirms how KANs improved the Transformer's ability to detect intrusions with fewer trainable parameters.

6. Conclusions and future works

At the end of this work, we can conclude that implementing Transformers using KAN layers shows promising performance in IDS within IoT environments rather than MLP layers. The TFKAN Transformer showed high accuracy in detecting intrusions across three recent different datasets. The proposed TFKAN Transformer outperforms and obtains higher accuracy scores of 99.96%, 98.43%, and 99.27 on the RT-IoT2022, IoT23, and CICIoT2023 datasets, respectively. The proposed TFKAN has trainable parameters less than the competitor by 75%, 80%, and 78% for the RT-IoT2022, IoT23, and CICIoT2023 datasets, respectively. The presented outcomes verify that the proposed TFKAN Transformer model has the proficiency to generalize all intrusion kinds and precisely acknowledge complicated routines in IDS by using fewer trainable parameters. So, Transformers based on KANs will be lighter by average 78% than Transformers based on MLPs, this makes KANs promising to be a replacement for MLPs. The training on the CICIoT2023 dataset with 46 million samples demonstrating that the model can scale to very large data volumes. One challenge to scalability is the increased training cost for KAN layers. However, this is mitigated by the reduced model size which aids deployment. Through the application of various hyper-parameter approaches, future research will expand on the optimization of TFKAN and minimize the processing time required during the training phase. Also, studying the applicability of KANs to situations involving real-time intrusion detection and prevention systems.

CRediT authorship contribution statement

Ibrahim A. Fares: Writing – original draft, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Mohamed Abd Elaziz:** Writing – review & editing, Visualization,

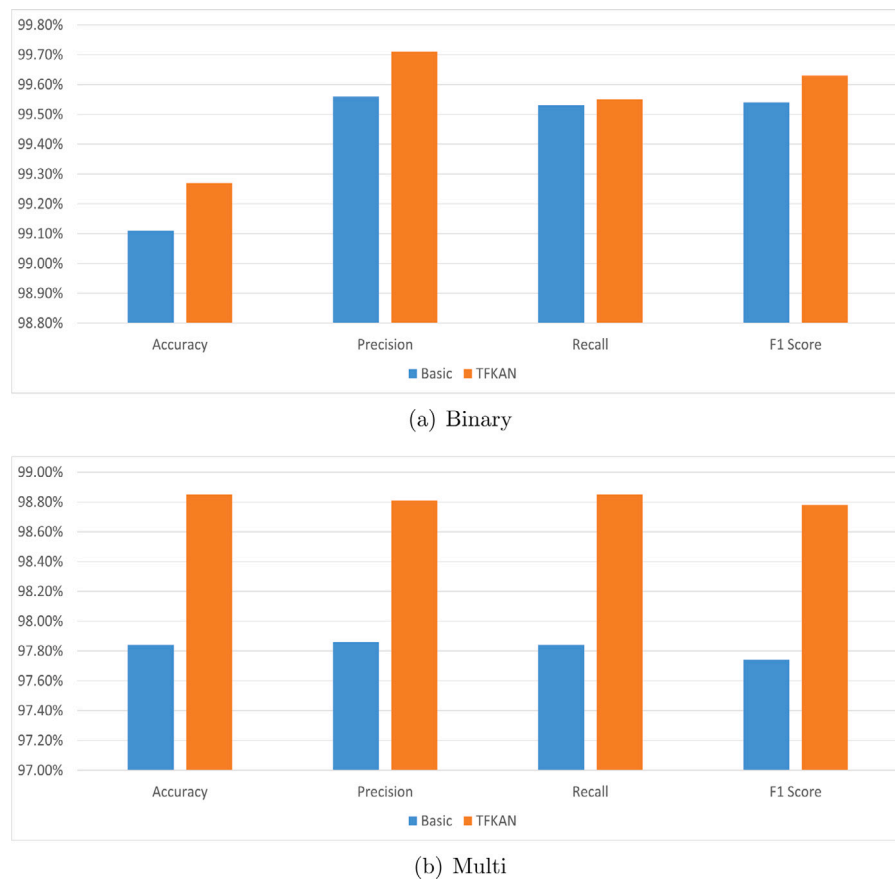


Fig. 9. The Bar chart of the Proposed TFKAN Transformer Compared to the Basic Transformer on CICIOT2023 Dataset.

Supervision, Software, Resources, Project administration. **Ahmad O. Aseeri**: Writing – review & editing, Investigation, Funding acquisition, Formal analysis, Conceptualization. **Hamed Shawky Zied**: Writing – review & editing, Visualization, Validation, Supervision, Software, Resources, Data curation, Conceptualization. **Ahmed G. Abdellatif**: Writing – review & editing, Supervision, Software, Resources, Methodology, Funding acquisition, Data curation, Conceptualization.

Declaration of competing interest

None.

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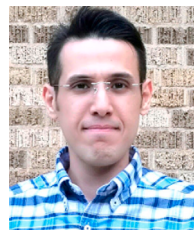
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